An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

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Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

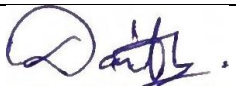
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

University students increasingly face challenges in managing academic workloads, maintaining productivity, and protecting their mental wellbeing. With multiple lectures, assignments, examinations, and sometimes part-time employment, many students struggle to organize their time effectively. Poor time management, excessive mobile phone usage, and insufficient sleep often lead to high stress levels and reduced academic performance. Existing productivity tools mainly focus on simple task management and do not intelligently analyze student behaviour or provide personalised academic planning based on mental wellbeing. This highlights a research gap in developing an intelligent system that integrates academic task management with behavioural analytics and stress prediction.

This research proposes an AI-driven academic productivity and wellbeing assistant that helps students plan their academic activities while monitoring their behavioural patterns. The system allows students to input weekly academic commitments such as lectures, assignments, examinations, and personal schedules. Using machine learning models, the system analyzes behavioural factors including study hours, sleep duration, phone usage, and task completion patterns to predict stress levels and productivity trends. Based on these predictions, the system automatically generates adaptive daily and weekly study plans that prioritise urgent academic tasks while maintaining a balanced workload to reduce burnout risk.

The proposed system will be implemented using technologies such as Python, machine learning libraries including Scikit-learn and XGBoost, and data processing tools like Pandas. The expected outcome is an intelligent decision-support system that improves student productivity, predicts stress levels, and promotes healthier study habits.

From a social perspective, the system can support student mental wellbeing and reduce academic burnout. Commercially, the solution has potential to be developed into a scalable educational technology application that can be adopted by universities and learning platforms.

Keywords: Academic Productivity, Student Stress Prediction, Machine Learning, Intelligent Scheduling, Mental Wellbeing

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
UI	User Interface
API	Application Programming Interface
REST	Representational State Transfer
DB	Database
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting
NLP	Natural Language Processing
SQL	Structured Query Language
KPI	Key Performance Indicator

INTRODUCTION

1.1 Background and Context

University students often face significant challenges in managing their academic responsibilities alongside personal activities. Assignments, examinations, group projects, lectures, and sometimes part-time jobs create a heavy workload that can be difficult to organize effectively. As a result, many students struggle with poor time management, missed deadlines, reduced productivity, and increasing levels of academic stress.

With the rapid growth of digital technology, many students use mobile applications and online tools to manage their schedules and tasks. However, most existing productivity tools mainly provide basic features such as task lists, reminders, or calendar management. These systems typically do not analyse student behaviour, learning patterns, or mental wellbeing. Therefore, they are limited in their ability to provide personalised guidance or adapt to the changing workload and stress levels of students.

Recent advancements in **Artificial Intelligence (AI)** and **Machine Learning (ML)** provide new opportunities to develop intelligent systems that can analyse user behaviour and generate adaptive recommendations. By analysing factors such as study time, sleep patterns, phone usage, task completion rates, and academic deadlines, AI models can identify patterns related to productivity, stress levels, and potential burnout risks.

In this context, there is a growing need for an intelligent academic support system that not only helps students organise their tasks but also monitors their productivity and mental wellbeing. Such a system can generate personalised study plans, predict stress levels, and dynamically adjust academic schedules based on student behaviour and workload.

Therefore, this research proposes the development of an **AI-driven intelligent assistant for academic productivity and mental wellbeing**, which integrates task scheduling, behavioural analysis, stress prediction, and adaptive planning. The goal of this system is to support university students in managing their academic workload more effectively while maintaining a healthy balance between productivity and mental wellbeing.

1.2 Literature Review

Academic stress and poor time management are significant issues affecting university students worldwide. Research shows that excessive academic workload, irregular study habits, and lifestyle factors such as sleep patterns and screen time can negatively affect student performance and mental wellbeing. Studies have shown that unmanaged stress can lead to anxiety, depression, and reduced academic productivity among university

students, highlighting the need for effective monitoring and support systems in educational environments.

With the rapid development of digital technology, many productivity tools have been introduced to help individuals organize tasks and schedules. However, most existing applications mainly provide basic functions such as task management, reminders, and calendar scheduling. These systems typically lack intelligent mechanisms that analyze behavioural patterns or mental health indicators. Therefore, they are limited in their ability to provide personalised recommendations or detect early signs of stress and burnout.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have enabled researchers to develop predictive models that analyse student behavioural data. Machine learning techniques have been increasingly used to predict academic stress levels by analysing factors such as study hours, sleep patterns, social media usage, and lifestyle habits. Several studies demonstrate that machine learning algorithms can effectively identify patterns related to stress and mental health among university students. For example, a study on student stress prediction applied supervised machine learning models such as Random Forest, Support Vector Machine (SVM), and Logistic Regression to classify stress levels among students. The results showed that some models achieved prediction accuracies of over 85–90%, demonstrating the effectiveness of machine learning techniques in analysing behavioural and academic data.

Similarly, another study developed a predictive model for identifying burnout risk among university students using decision tree algorithms and psychological variables. The research highlighted that factors such as workload, academic pressure, and lifestyle behaviours significantly contribute to student burnout. Early detection using predictive models can help educational institutions provide timely interventions and support services.

Recent research has also explored the use of behavioural and sensor data to predict student stress levels. For example, studies have analysed mobility patterns, lifestyle habits, and physiological data using machine learning pipelines to identify relationships between behavioural patterns and stress levels among university students. These approaches demonstrate that data-driven methods can be used to monitor student wellbeing and support mental health interventions.

Despite these advancements, many existing studies focus mainly on stress prediction or academic performance prediction separately. Few systems integrate task management, behavioural analysis, mental health prediction, and adaptive scheduling into a single intelligent platform. Therefore, there is a research gap in developing an integrated AI-based academic assistant that can monitor student behaviour, predict stress levels, and dynamically generate personalised study schedules.

This research aims to address this gap by developing an **AI-driven intelligent assistant for academic productivity and mental wellbeing**, which integrates machine learning-based stress prediction, productivity analysis, and adaptive scheduling to support university students in managing their academic workload effectively.

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1.3 Research Gap

Although many studies and applications have focused on improving student productivity and mental health monitoring, several limitations still exist in current systems. Many existing productivity tools mainly provide basic features such as task scheduling, reminders, and calendar management. For example, applications like Google Calendar and Todoist help users organize tasks and deadlines, but they do not analyse student behavioural patterns or mental wellbeing.

In addition, several research studies have used Artificial Intelligence and Machine Learning techniques to predict student stress levels or academic performance. However, most of these studies focus only on a **single aspect**, such as stress prediction or academic performance analysis, without integrating task management, behavioural monitoring, and adaptive planning into one system.

Another limitation is that many existing approaches rely mainly on **static data collected through surveys** rather than continuous behavioural data such as study hours, sleep patterns, phone usage, and task completion progress. As a result, these systems are not capable of dynamically adapting study schedules or providing real-time recommendations based on the student's daily activities and workload.

Furthermore, current productivity applications rarely include mechanisms for **early burnout detection or adaptive scheduling** that can automatically adjust workloads based on predicted stress or productivity levels. This makes it difficult for students to maintain a balanced academic routine while managing multiple responsibilities.

Therefore, there is a clear need for an **integrated AI-driven system** that combines academic task management, behavioural data analysis, stress prediction, and adaptive

scheduling in a single platform. This research aims to address this gap by developing an intelligent assistant that monitors student behaviour, predicts stress and productivity levels using machine learning techniques, and dynamically generates personalised study plans to support both academic productivity and mental wellbeing.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to **develop an AI-driven intelligent assistant that helps university students manage their academic tasks efficiently while monitoring their productivity and mental wellbeing**.

The proposed system will analyse students' academic schedules, behavioural patterns, and daily activities to **predict stress levels, evaluate productivity, and generate adaptive study plans**. Using **Artificial Intelligence and Machine Learning techniques**, the system will dynamically adjust task schedules based on deadlines, workload, and user behaviour to help students improve time management, maintain productivity, and reduce academic stress.

2.2 Specific Objectives

1. **To design and develop a system that allows students to input academic activities** such as lectures, assignments, exams, and work schedules in order to organise their weekly tasks.

2.**To develop a task scheduling mechanism** that automatically generates daily study plans based on deadlines, task priorities, and available free time.

3. **To analyse student behavioural data** such as study hours, sleep patterns, phone usage, and task completion rates to understand productivity patterns.

4. **To develop Machine Learning models to predict student stress levels and productivity levels** using behavioural and academic activity data.

5. **To implement an adaptive scheduling system** that dynamically adjusts weekly study plans based on predicted stress levels, productivity patterns, and academic workload.

6. **To provide personalised recommendations and motivational feedback** to help students maintain a balance between academic productivity and mental wellbeing.

3.METHODOLOGY

This research will follow a **data-driven system development approach** that integrates task management, behavioural monitoring, and machine learning techniques to support academic productivity and mental wellbeing. The proposed system will collect academic and behavioural data from students, analyse the data using machine learning models, and generate adaptive study schedules based on predicted productivity and stress levels.

The research methodology consists of several stages including **data collection, data preprocessing, machine learning model development, system development, and evaluation**.

3.1 Research Approach

This research follows a **Design Science Research (DSR)** approach within the field of Artificial Intelligence and Machine Learning. The objective is to design and develop an intelligent system that assists university students in managing their academic workload while monitoring their productivity and stress levels.

The study involves designing an AI-based system, implementing predictive models, and evaluating their effectiveness using real or publicly available student behavioural datasets. The approach includes problem identification, system design, model development, experimental evaluation, and performance analysis.

3.2 Technical Methods and Algorithms

The proposed system uses several machine learning techniques to analyse student behavioural patterns and generate adaptive study plans.

Stress Prediction Model:
A classification model will be used to predict student stress levels based on behavioural indicators such as sleep hours, study hours, phone usage, and task completion rate. Algorithms such as **Random Forest** and **Logistic Regression** will be implemented using the Scikit-learn framework.

Productivity Prediction Model:
A regression model will estimate the student's productivity score using behavioural features including study duration, lecture attendance, and phone usage patterns. The gradient boosting algorithm implemented through XGBoost will be used due to its high performance with structured datasets.

Task Prioritisation Algorithm:
A priority-based scheduling algorithm will generate daily to-do lists by analysing assignment deadlines, estimated workload, and available study time.

Burnout Risk Detection:
A classification model will analyse trends in stress levels, sleep patterns, and workload to detect potential burnout risk.

3.3 Tools and Platforms with Justification

The system will be developed using the following tools:

- **Python** programming language for implementing machine learning models due to its strong ecosystem for data analysis and AI development.
- Pandas and NumPy for data preprocessing and feature engineering.
- Scikit-learn and XGBoost for machine learning model development.
- Firebase for storing user activity data and managing application data.
- A mobile or web interface developed using modern frameworks to allow students to input schedules and track their progress.

These tools were selected because they provide reliable performance, strong community support, and efficient implementation of machine learning algorithms.

3.4 Data Collection and ethical consideration

The research will use publicly available datasets related to student productivity, academic stress, and mental wellbeing. These datasets contain anonymised behavioural data such as study hours, sleep patterns, and social media usage.

If user testing is conducted, participation will be voluntary and participants will be informed about the purpose of the study. Personal information will not be collected, and all collected data will remain anonymous and confidential. The system will only analyse behavioural patterns related to academic activities and will not access sensitive personal information.

3.5 Validation Strategy and Performance Metrics

The developed models will be evaluated using standard machine learning validation techniques such as **train-test splitting** and **cross-validation**.

For classification models (stress prediction and burnout detection), the following performance metrics will be used:

- Accuracy
- Precision
- Recall
- F1-Score

For regression models (productivity prediction), the evaluation metrics include:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Additionally, the effectiveness of the scheduling system will be evaluated by analysing improvements in task completion rates and workload balance. These evaluations will help determine the reliability and practical usefulness of the proposed AI-based academic productivity assistant.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

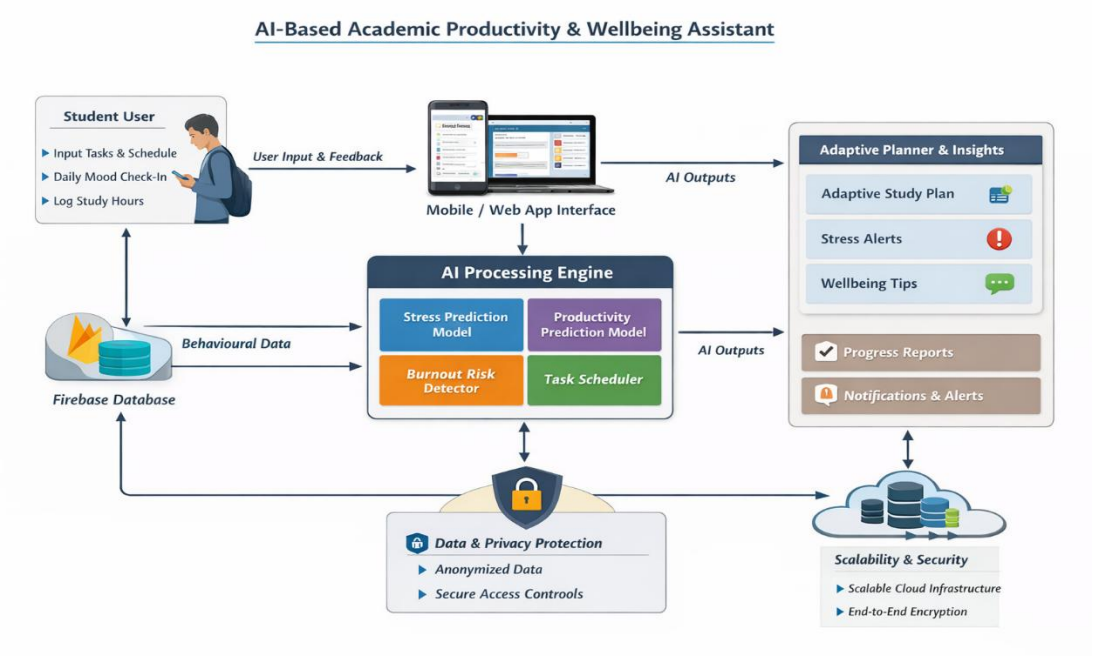


Figure 1 System Architecture

5. USER REQUIREMENTS

The proposed AI-based academic productivity assistant involves several stakeholders who interact with or benefit from the system.

5.1 Stakeholder Identification.

Primary Stakeholders

1. **University Students**
   The main users of the system. Students will use the application to input academic schedules, track study tasks, monitor stress levels, and receive personalised study recommendations.

2. **Lecturers / Academic Advisors**
   They may use aggregated reports to understand students' academic workload patterns and identify students who may require academic support.

Secondary Stakeholders

3. **University Administration**
   Institutions may use insights from the system to improve academic support services and student wellbeing programs.

4. **Researchers and Developers**
   They may use the collected anonymised data to further improve AI models and study student productivity behaviours in the field of Artificial Intelligence and Machine Learning.

5.2 Functional Requirements

The system must provide the following core functionalities:

FR1: The system shall allow students to create user accounts and log into the application securely.

FR2: The system shall allow students to input their weekly academic schedule, including lectures, assignments, examinations, and deadlines.

FR3: The system shall automatically generate daily to-do lists based on the provided schedule and task priorities.

FR4: The system shall allow students to track task completion status (Completed, Partial, or Skipped).

FR5: The system shall allow students to record daily behavioural data such as study hours, sleep duration, and mood check-ins.

FR6: The system shall analyse student behaviour using machine learning models to predict stress levels and productivity patterns.

FR7: The system shall detect potential burnout risk based on workload and behavioural trends.

FR8: The system shall generate adaptive weekly study plans based on previous performance and predictions.

FR9: The system shall provide notifications and reminders for upcoming deadlines and planned study sessions.

FR10: The system shall generate visual reports and insights on student productivity, task completion rate, and stress trends.

5.3 Non-Functional Requirements

The system must also satisfy several quality attributes to ensure reliability and usability.

Security

- User data must be securely stored with authentication mechanisms.
- Sensitive behavioural data must be protected and anonymised where necessary.
- Secure communication protocols must be used to protect data transmission.

Performance

- The system should generate daily task recommendations within a few seconds.
- Machine learning predictions should be processed efficiently without noticeable delays for the user.

Usability

- The application interface should be simple and easy to use for university students.
- Users should be able to complete daily task updates within a few seconds.

Reliability

- The system should ensure consistent operation with minimal downtime.
- User data should be regularly backed up to prevent data loss.

Scalability

- The system architecture should support an increasing number of users without performance degradation.

Privacy

- Personal student information must not be shared with third parties without user consent.
- All research data used for analysis should be anonymised.

6. COMMERCIALIZATION PLAN

The proposed intelligent assistant for academic productivity and mental wellbeing has strong potential for commercialization due to the increasing demand for digital tools that support student productivity and mental health management. The commercialization strategy focuses on transforming the developed research prototype into a scalable and market-ready application for students and educational institutions.

6.1 Target Market and Customer Persona

Target Market

The primary target market for the proposed intelligent assistant system is university and college students who face challenges in managing academic workloads, maintaining productivity, and balancing their mental wellbeing. Many students struggle with organizing assignments, preparing for examinations, and managing multiple responsibilities such as part-time jobs and extracurricular activities. The proposed system aims to support these students by providing an AI-driven platform that helps them plan their schedules, monitor productivity, and manage stress levels.

The potential target market includes:

- Undergraduate students who need assistance in managing coursework, assignments, and exams.
- Postgraduate students who often manage research work, academic deadlines, and teaching responsibilities.
- Students with part-time jobs who need better time management tools to balance work and studies.
- Universities and educational institutions that aim to improve student productivity and mental wellbeing through digital support systems.

This system can also be adopted by universities as part of their student support and academic success programs, allowing institutions to promote better study habits and mental health awareness among students.

Customer Persona

A customer persona represents a typical user who would benefit from the proposed system.

Persona Example: University Student

Name: Nimal Perera
Age: 21
Occupation: Undergraduate Student
Education Level: Third-year university student
Lifestyle: Attends lectures, works on assignments and group projects, and works part-time during weekends.

Challenges:

- Difficulty managing multiple academic deadlines.
- Poor time management and inconsistent study routines.
- Increased stress during examination periods.
- Frequent distractions from mobile phone and social media usage.

Needs:

- A system that helps organize academic schedules.
- A tool that automatically generates daily study plans.
- Insights into productivity and stress levels.
- Reminders and motivational support to stay focused.

How the System Helps:
The proposed intelligent assistant helps the student manage academic tasks more effectively by generating personalized study schedules, tracking productivity patterns, predicting stress levels, and providing recommendations to maintain a balanced academic routine.

6.2 Value Proposition

The proposed system offers a unique value by combining **academic task management, behavioural analysis, and Artificial Intelligence–based predictions** into a single intelligent platform designed specifically for university students. Unlike traditional productivity tools that only provide basic task lists or reminders, this system uses **Machine Learning techniques to analyse student behaviour and generate personalised study plans** that adapt to changing workloads and stress levels.

The main value of the proposed system lies in its ability to **improve student productivity while supporting mental wellbeing**. By analysing factors such as study hours, sleep patterns, task completion rates, and academic deadlines, the system can predict productivity and stress levels and adjust study schedules accordingly. This helps students manage their time more effectively, reduce academic pressure, and maintain a balanced study routine.

In addition, the system provides **real-time insights and recommendations** that guide students toward healthier academic habits. Features such as adaptive scheduling, workload balancing, and motivational feedback encourage students to stay organized and focused throughout the academic term.

For educational institutions, the system provides value by helping improve **student academic performance and overall wellbeing**. Universities can benefit from better student engagement, improved time management skills, and reduced academic stress among students.

Overall, the proposed intelligent assistant delivers value by **integrating AI-driven productivity management with mental wellbeing monitoring**, creating a comprehensive solution that supports students in achieving both academic success and personal balance.

6.3 Revenue Model

The proposed intelligent assistant for academic productivity and mental wellbeing can adopt several revenue generation strategies to ensure long-term sustainability and scalability. The revenue model focuses on providing value to students and educational institutions while maintaining accessibility for users.

Freemium Model

The system can follow a **freemium business model**, where basic features are provided free of charge to attract a large number of student users. These features may include task scheduling, deadline reminders, and basic progress tracking. Advanced features

such as AI-based productivity analysis, personalised study recommendations, and detailed progress reports can be offered through a premium subscription plan.

Subscription-Based Model

Students who require additional features can subscribe to a **monthly or yearly premium plan**. Premium services may include advanced analytics dashboards, stress prediction insights, adaptive scheduling improvements, and personalized productivity reports.

Institutional Licensing

Educational institutions such as universities and colleges can purchase **institutional licenses** that provide access to the platform for their students. Universities may integrate the system into their academic support or student wellbeing programs. Institutional adoption can provide a stable revenue stream while supporting large-scale student usage.

Partnership Opportunities

The platform may also collaborate with **educational service providers, online learning platforms, and student support organizations**. These partnerships can enable integration with learning management systems and create opportunities for co-developed features or sponsored services.

Future Monetization Opportunities

In the future, the system may generate revenue through additional services such as:

- Advanced analytics tools for educational institutions
- Personalized academic performance insights
- Integration with career development and skill-building platforms

6.4 Cost Estimation

The estimated costs for developing and maintaining the proposed intelligent assistant system include development costs and operational costs. These costs cover the resources required for system implementation, deployment, and ongoing maintenance.

Development Costs

The development phase includes designing, building, and testing the intelligent assistant system. The main development cost components include:

- **Mobile Application Development**
  Development of the user interface and mobile application that allows students

to input academic tasks, track progress, and receive AI-generated recommendations.

- **Backend Infrastructure Development**
  Implementation of backend services, APIs, and database management systems that support data storage, system communication, and application functionality.
- **Machine Learning Model Development**
  Development and training of machine learning models used for predicting student productivity and stress levels, as well as generating adaptive study plans.

Operational Costs

Operational costs refer to the expenses required to maintain and operate the system after deployment.

- **Cloud Hosting and Database Services**
  Cloud platforms will be used to store user data, run backend services, and support system scalability.
- **Server Maintenance and Updates**
  Regular system updates and maintenance will be required to ensure stable system performance and improve the machine learning models.
- **Technical Support**
  Ongoing technical support may be required to handle system issues, user feedback, and feature improvements.

Cost Optimization

Using cloud platforms such as **Firebase** helps reduce infrastructure costs during the early development stages. Cloud-based services provide scalable storage, authentication services, and real-time databases without requiring expensive physical server infrastructure. This approach allows the system to operate efficiently while minimizing initial investment costs.

6.5 Pricing Strategy

The proposed pricing strategy is designed to encourage student adoption while ensuring the long-term sustainability of the system. A **freemium pricing model** will be used so that users can access basic features for free, while advanced features are available through a premium subscription.

Example Pricing Plan

- **Free Version:**
  Provides basic task management, deadline reminders, and study scheduling features.
- **Premium Version:**
  Provides advanced features such as productivity analytics, burnout prediction, adaptive study scheduling, and detailed progress reports.

Subscription Options

Possible subscription plans may include:

- **Monthly Plan:**
  A low-cost monthly subscription designed to be affordable for university students.
- **Annual Plan:**
  A discounted yearly subscription that provides full access to premium features at a lower overall cost.

This pricing strategy encourages students to adopt the platform through free basic features while providing the option to upgrade for more advanced AI-powered capabilities.

6.6 Competitive Advantage

The proposed intelligent assistant offers several competitive advantages compared to traditional productivity and task management applications. While many existing tools provide basic task scheduling and reminder features, the proposed system integrates **Artificial Intelligence and Machine Learning techniques** to provide more intelligent and personalized support for students.

AI-Driven Personalization

Unlike standard productivity applications that rely on manual scheduling, the proposed system uses machine learning models to analyse student behaviour and generate personalised study plans. This allows the system to adapt schedules based on productivity patterns, workload, and stress levels.

Integrated Productivity and Wellbeing Monitoring

Most existing productivity tools focus only on task management. In contrast, the proposed system combines **academic task management with mental wellbeing monitoring**, allowing the system to predict stress levels and detect early signs of burnout.

Adaptive Scheduling

The system dynamically adjusts study schedules based on deadlines, task completion rates, and predicted productivity levels. This adaptive approach helps students manage their workload more effectively and reduce academic stress.

Data-Driven Insights

The system provides users with insights into their productivity trends, task completion patterns, and stress levels. These insights help students understand their study habits and make better decisions regarding time management.

Student-Focused Design

The system is specifically designed for university students, taking into account academic schedules, assignment deadlines, exam preparation, and student lifestyle factors. This makes the platform more relevant and useful compared to general productivity tools.

6.7 Intellectual Property Considerations

Intellectual Property (IP) considerations are important to ensure that the innovations, software components, and research outcomes developed in this project are properly protected and managed. The proposed intelligent assistant system involves the development of software algorithms, machine learning models, and system architecture that may be considered intellectual property.

Software Ownership

The source code, algorithms, and system architecture developed during this research will be considered the intellectual property of the researcher and the affiliated academic institution. Proper documentation and version control will be maintained to record the development process and ownership of the system components.

Use of Open-Source Technologies

The proposed system will utilize several open-source technologies and libraries, such as Python-based machine learning frameworks and development tools. These tools are widely used for research and development and will be used in compliance with their respective open-source licenses. Proper attribution will be provided where required.

Protection of Innovative Components

Certain components of the system, such as the adaptive scheduling algorithm, behavioural analysis model, or system architecture, may represent novel research contributions. These innovations may be eligible for intellectual property protection,

such as software copyright or potential patent applications, depending on institutional policies and the level of novelty.

Data Privacy and Ownership

User data collected by the system, including academic schedules and behavioural data, will be handled with strict privacy considerations. Data will be used only for research and system improvement purposes, and appropriate measures will be implemented to ensure confidentiality and secure storage.

Future Commercialization Rights

If the system is further developed into a commercial product, intellectual property ownership and commercialization rights will be managed according to the policies of the affiliated academic institution and relevant legal regulations.

7.BUDGET AND JUSTIFICATION

The budget for this research project includes the estimated costs required for system development, cloud services, and research implementation. The selected tools and technologies are mostly open-source to minimize development costs while maintaining efficiency and scalability.

7.1 Hardware Costs

Item	Description	Estimated Cost (LKR)
Laptop / Development Computer	Used for system development, machine learning model training, and testing	LKR 250,000
Internet Connectivity	Required for development, dataset access, and cloud services during the research period	16,000

Table 1Hardware Costs

Subtotal (Hardware): 266,000 LKR

7.2 Software Licenses

Software	Purpose	Estimated Cost (LKR)
Python	Programming language for AI and system development	Free
Scikit-learn	Machine learning model implementation	Free
XGBoost	Advanced machine learning model development	Free
Visual Studio Code	Development environment	Free
GitHub	Version control and project management	Free

Table 2 Software Licenses

Subtotal (Software): 0 LKR

7.3 Cloud Services

Service	Purpose	Estimated Cost (LKR)
Firebase Cloud Database	Data storage, authentication, and backend services	LKR 10,000
Cloud Hosting	Deployment of backend services and APIs	LKR15,000

Table 3 Cloud Services

Subtotal (Cloud Services): 25,000 LKR

7.4 Human Effort Cost

Role	Description	Estimated Cost (LKR)
Researcher / Developer	Time spent on system design, development, machine learning model training, and testing	LKR 50,000

Table 4 Human Effort Cost

Subtotal (Cloud Services): 50,000 LKR

7.5 Total Estimated Budget

Category	Cost (LKR)
Hardware	LKR 266,000
Software	LKR 0
Cloud Services	LKR 25,000
Human Effort	LKR 50,000

Table 5 Total Estimates

Total Estimated Budget: 341,000 LKR

Budget Justification Summary

The hardware cost includes a development computer that is necessary for implementing the system and training machine learning models. Reliable internet connectivity is required for accessing datasets, cloud services, and development resources.

Most of the software tools used in the project are open-source, which significantly reduces licensing costs while providing powerful development and machine learning capabilities.

Cloud services are required to host the backend system, store user data, and support system scalability. Using cloud platforms helps reduce infrastructure costs and

eliminates the need for maintaining physical servers during the early stages of development.

The human effort cost represents the time and expertise required for designing the system architecture, developing machine learning models, implementing the application, and evaluating the system performance.

8.WORK BREAKDOWN STRUCTURE (WBS)

Since **you are the only developer/researcher responsible for this component**, the tasks are organized into phases across an **8-month development timeline**. The work breakdown ensures a clear sequence of research, development, implementation, and evaluation activities.

8.1 Task Breakdown

Task ID	Task	Description
T1	Literature Review	Study existing research on productivity systems, stress monitoring, and burnout prediction.
T2	Problem Definition	Define research problem, objectives, and system requirements.
T3	System Design	Design system architecture, database structure, and UI wireframes.
T4	Dataset Preparation	Collect and preprocess datasets related to stress, workload, and productivity
T5	Machine Learning Model Development	Train and test ML models for stress or burnout prediction
T6	Mobile Application Development	Develop React Native application for task input and user interaction
T7	Backend & Database Integration	Implement Firebase backend for data storage and synchronization
T8	Adaptive Scheduling Algorithm	Implement task prioritization and workload balancing algorithm

T9	Dashboard Development	Create productivity and wellbeing insights dashboard
T10	System Integration	Integrate mobile app, backend, and machine learning components
T11	System Testing & Evaluation	Test system performance, usability, and model accuracy
T12	Documentation & Final Report	Prepare research documentation, thesis report, and final presentation

Table 6 Task Breakdown

8.2 Timeline Plan

Month	Activities
Month 1	Literature review, problem identification, research proposal preparation
Month 2	System requirement analysis and architecture design
Month 3	Dataset collection and preprocessing
Month 4	Machine learning model development and testing
Month 5	Mobile application development (React Native)
Month 6	Backend integration and adaptive scheduling algorithm implementation
Month 7	System integration, dashboard development, and testing
Month 8	System evaluation, performance analysis, documentation, and final report preparation

Table 7 Timeline Plan

8.3 Workload Management

Since the project is carried out by a single researcher, tasks are organized sequentially to ensure a manageable workload and smooth progress throughout the research period. The project activities are divided into several phases, allowing each stage of the research to be completed systematically.

Initial activities such as **literature review, requirement analysis, and system design** are completed in the early phase of the project. These tasks establish the foundation for the development of the proposed system.

The next phase focuses on **data collection, preprocessing, and machine learning model development**. During this stage, models for predicting student productivity and stress levels are developed and evaluated.

Following this, the project moves to **system development and integration**, including mobile or web application development, backend implementation, and integration of machine learning models into the system.

Finally, the project concludes with **system testing, evaluation, and documentation** to ensure the system functions correctly and the research findings are properly recorded.

This structured approach ensures that the proposed intelligent assistant system can be **designed, implemented, tested, and evaluated within the planned 8-month research timeline** while maintaining a balanced workload for the researcher.

9.GANTT CHART

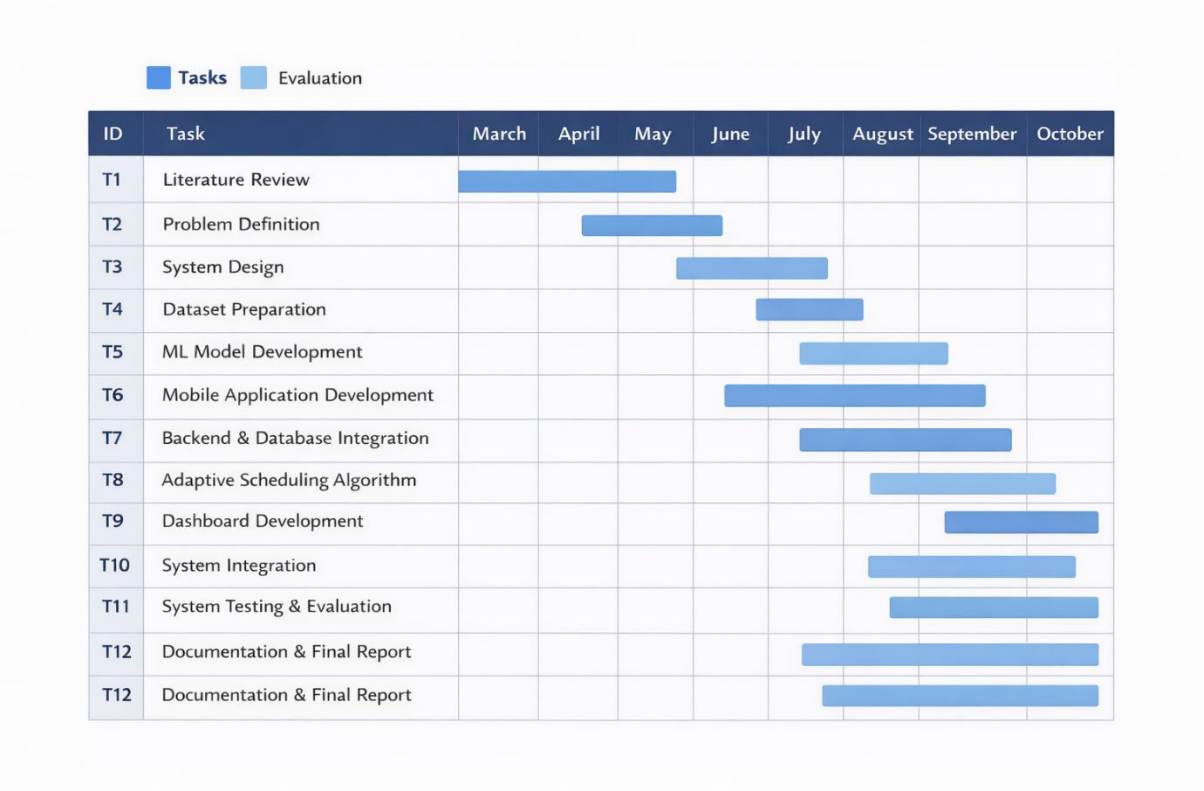


Figure 2 Grantt Chart

10. REFERENCES LIST

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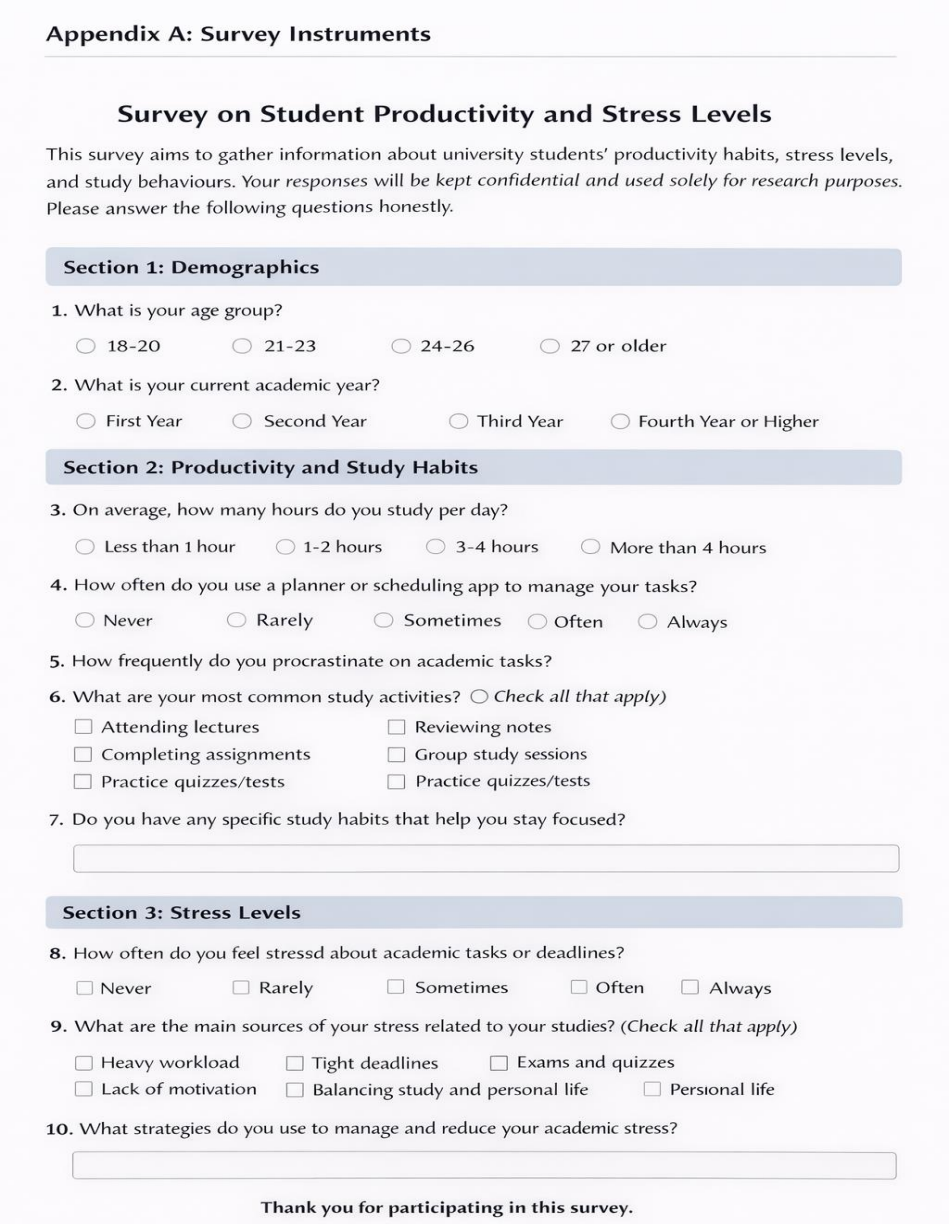
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11. APPENDICES

11.1 Survey instruments



Appendix A: Survey Instruments

Survey on Student Productivity and Stress Levels

This survey aims to gather information about university students' productivity habits, stress levels, and study behaviours. Your *responses will be kept confidential and used solely for research purposes.* Please answer the following questions honestly.

Section 1: Demographics

1. What is your age group?
   ○ 18-20 ○ 21-23 ○ 24-26 ○ 27 or older
2. What is your current academic year?
   ○ First Year ○ Second Year ○ Third Year ○ Fourth Year or Higher

Section 2: Productivity and Study Habits

3. On average, how many hours do you study per day?
   ○ Less than 1 hour ○ 1-2 hours ○ 3-4 hours ○ More than 4 hours
4. How often do you use a planner or scheduling app to manage your tasks?
   ○ Never ○ Rarely ○ Sometimes ○ Often ○ Always
5. How frequently do you procrastinate on academic tasks?
6. What are your most common study activities? ○ *Check all that apply)*
   ☐ Attending lectures ☐ Reviewing notes
   ☐ Completing assignments ☐ Group study sessions
   ☐ Practice quizzes/tests ☐ Practice quizzes/tests
7. Do you have any specific study habits that help you stay focused?

Section 3: Stress Levels

8. How often do you feel stressd about academic tasks or deadlines?
   ☐ Never ☐ Rarely ☐ Sometimes ☐ Often ☐ Always
9. What are the main sources of your stress related to your studies? *(Check all that apply)*
   ☐ Heavy workload ☐ Tight deadlines ☐ Exams and quizzes
   ☐ Lack of motivation ☐ Balancing study and personal life ☐ Persional life
10. What strategies do you use to manage and reduce your academic stress?

Thank you for participating in this survey.

11.2 Detailed diagrams

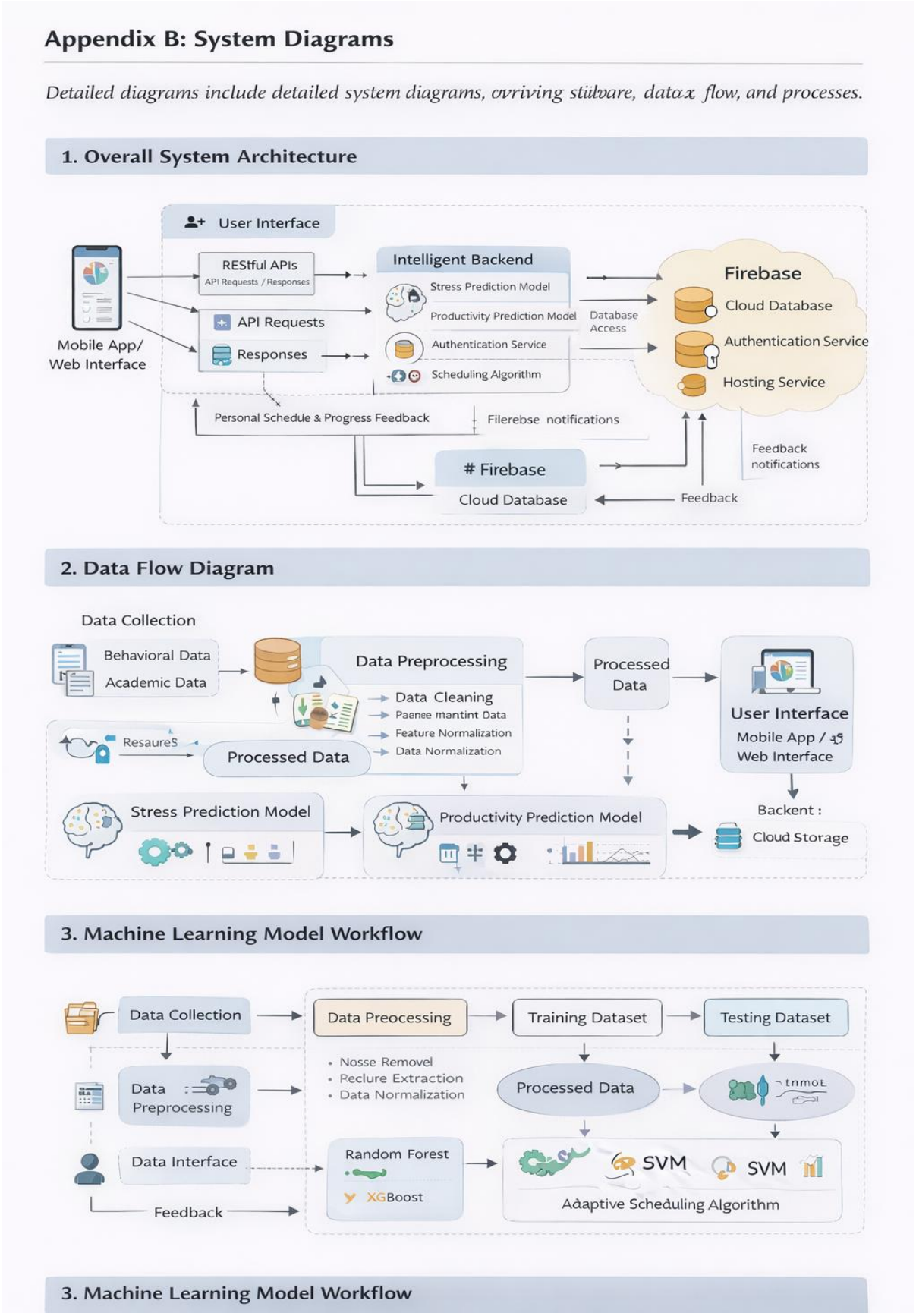


Figure 3 Detailed diagram

11.3 Preliminary UI mockups

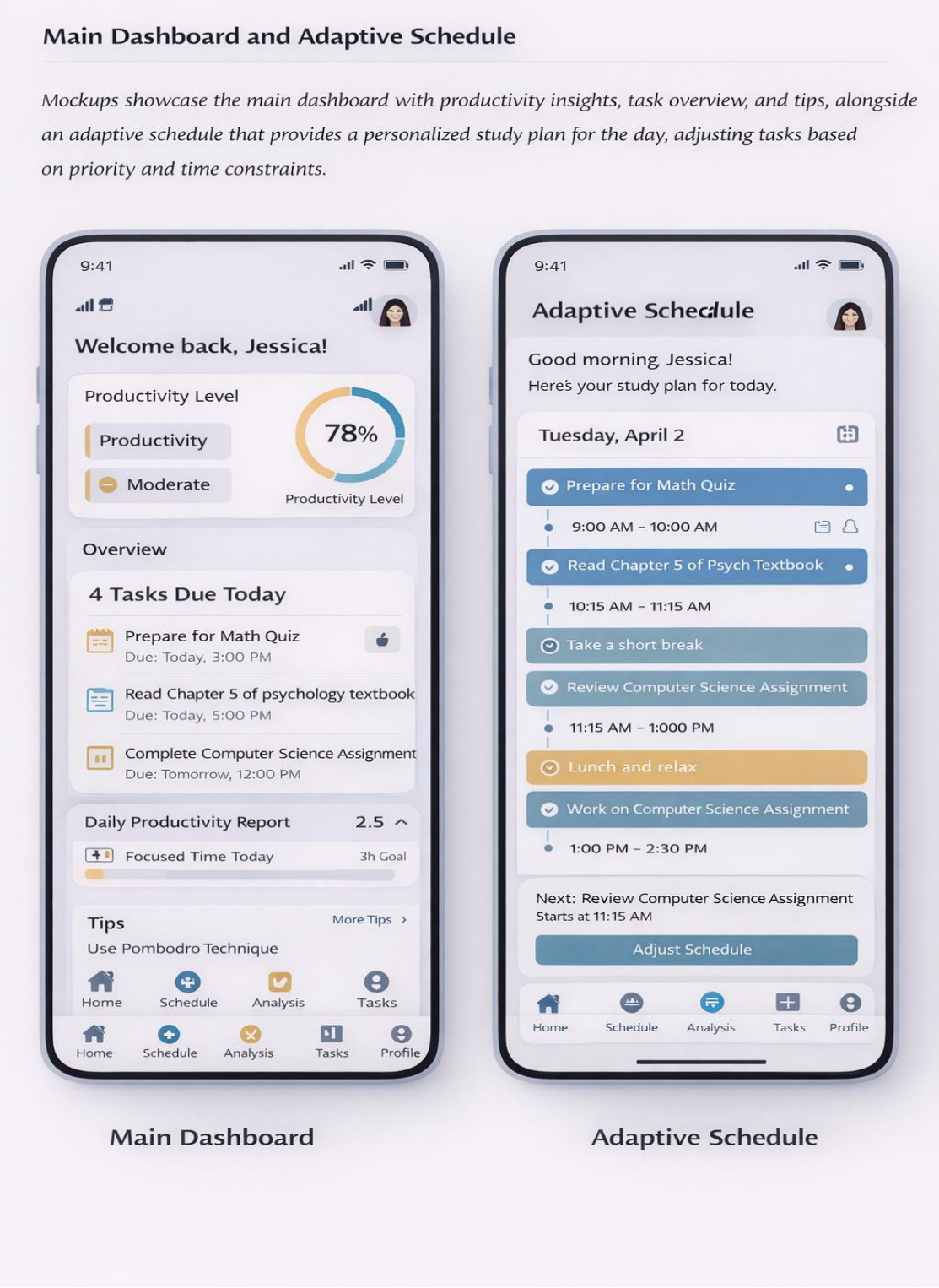


Figure 4 Preliminary UI mockups 1

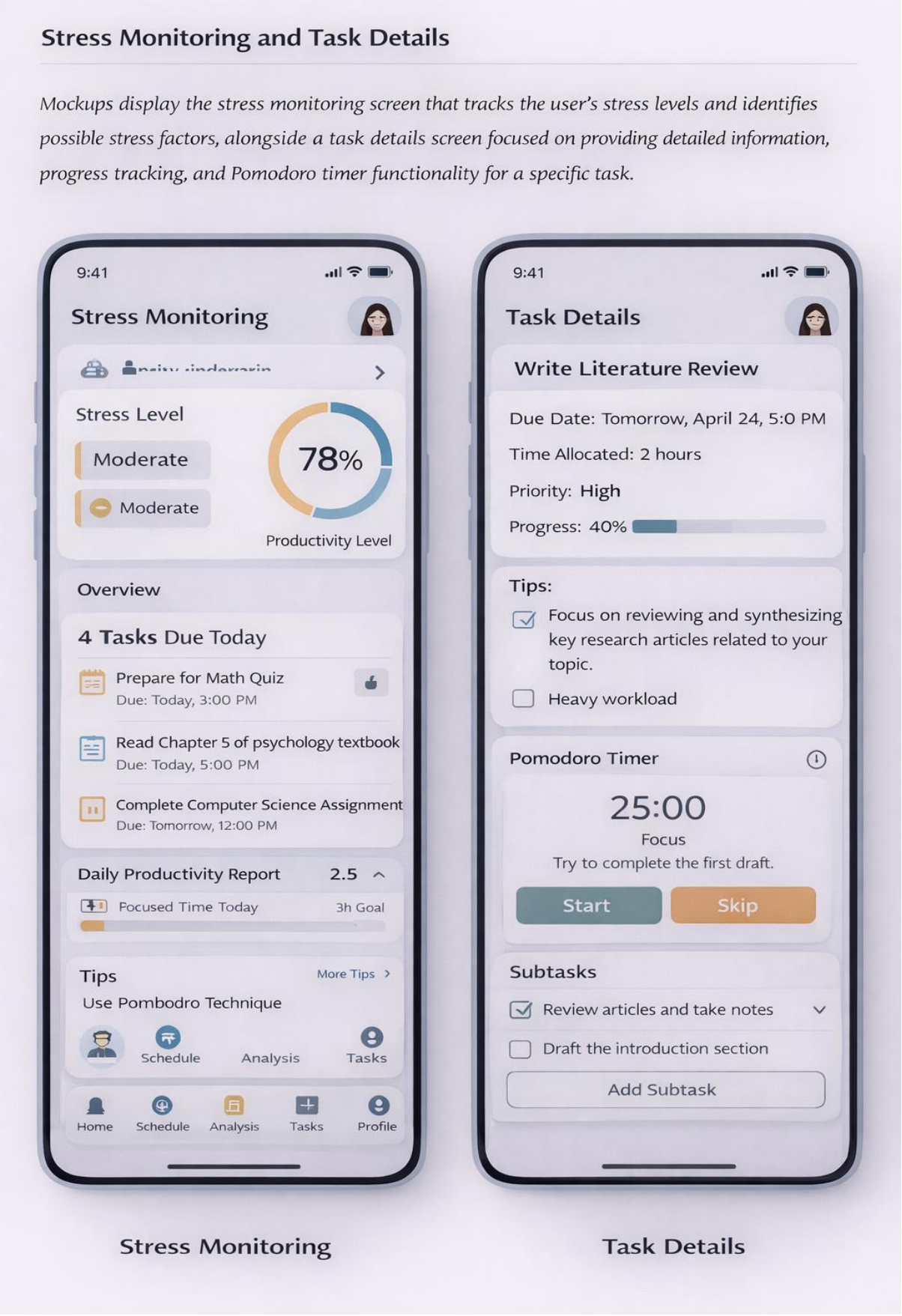


Figure 5 Preliminary UI mockups 2

11.4 Risk analysis

Risk analysis identifies potential challenges that may affect the successful development and implementation of the proposed intelligent assistant system. The following table outlines possible risks and the strategies that will be used to mitigate them.

Risk	Description	Mitigation Strategy
Data Quality Issues	Collected datasets may contain missing values, noise, or inconsistent data that could affect machine learning model accuracy.	Apply proper data preprocessing techniques such as data cleaning, normalization, and feature selection.
Machine Learning Model Accuracy	The predictive models for stress and productivity may not achieve the expected accuracy.	Test multiple algorithms and tune model parameters to improve prediction performance.
Data Privacy Concerns	The system collects behavioural data such as study hours and stress levels, which may raise privacy concerns.	Ensure secure data storage, anonymize user data, and follow ethical research guidelines.
Technical Integration Challenges	Integrating machine learning models with the mobile or web application may introduce technical difficulties.	Use modular system design and well-documented APIs to simplify integration.
Time Constraints	The research project must be completed within the planned 8-month timeline.	Follow a structured project schedule and complete tasks according to the defined work breakdown structure.
System Performance Issues	The application may experience performance delays when processing large datasets.	Optimize data processing techniques and use cloud services for scalable computing resources.

11.5 Ethical Documents

This research involves collecting behavioural and academic-related information from students to analyse productivity patterns and stress levels. Therefore, ethical considerations are important to ensure that the research is conducted responsibly and that participant data is protected.

Informed Consent Form

Participants involved in surveys or data collection will be provided with an informed consent form. This document will clearly explain the purpose of the research, the type of data being collected, and how the data will be used. Participants will be required to voluntarily agree to participate before providing any information.

Participant Information Sheet

A participant information sheet will be provided to explain the objectives of the research, the procedures involved, and the expected duration of participation. It will also inform participants that their involvement is voluntary and that they have the right to withdraw from the study at any time without any consequences.

Data Privacy and Confidentiality Agreement

All collected data will be treated as confidential and will be used only for research purposes. Personal information will be anonymized and securely stored to ensure participant privacy. Access to the collected data will be restricted to the researcher.

Ethical Approval

If required by the academic institution, the research proposal will be submitted to the relevant ethical review committee for approval before conducting surveys or collecting participant data. This ensures that the research follows proper ethical standards and guidelines.